

Scenario C Explanation — Fairness and Starvation-Sensitive Case:

Scenario C explores fairness and potential starvation. A long-burst, low-priority process (P1) arrives first, while several short-burst, high-priority processes (P2, P3, P4) arrive shortly after. Under Priority Scheduling, P1 is forced to wait until all higher-priority processes complete, causing a very long waiting time and demonstrating the starvation problem. Under SJF, P1 runs first because it is the only process available at time 0, after which the shorter jobs finish quickly.

Process	Arrival Time	Burst Time	Priority
P1	0	2	5
P2	1	10	1 (highest)
P3	1	4	3
P4	2	1	4

SJF Execution Order: P1 → P4 → P3 → P2 (0–2 → 2–3 → 3–7 → 7–17)

Priority Execution Order: P1 → P2 → P3 → P4 → P2 (see Gantt) → ... finishing at time 17

SJF Average: WT = 2.25 | TAT = 6.50 | RT = 2.25

Priority Average: WT = 9.00 | TAT = 13.25 | RT = 9.00

Key Insight: Under Priority Scheduling, process P1 (low priority = 5) is forced to wait until time 15 even though it arrived first and has one of the shortest burst times. This is a clear demonstration of starvation — a process that is ready and short can be indefinitely delayed by a stream of higher-priority arrivals. SJF, by contrast, naturally favors short jobs and produces a much lower average waiting time (2.25 vs 9.00), illustrating that the two algorithms optimize for fundamentally different goals.